

## Chapter 4

# Inception is NOT the Requirement Phase

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# What is Inception?

- Most projects require a short initial step to question about:
  - What is the vision and business case for this project?
  - Feasible? *↳ define the scope.*
  - Buy and/or build?
  - Rough unreliable range of cost: Is it \$10K-100K or in the millions?
  - Should we proceed or stop?

- Inception should be short.

- One week for most projects. No iterations!
- Most requirements analysis occurs during the elaboration phase, not inception.

*↳ Inception은 요구사항 분석이 아니다. analysis 단계가 아니다.*



# Artifacts Start in Inception

| Artifact                         | Comment                                                                                                                                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Vision and Business Case         | Describes the high-level goals and constraints, the business case, and provides an executive summary.                                                                                                                 |
| Use-Case Model                   | Describes the <u>functional requirements</u> . During inception, <b>the names of most use cases will be identified</b> , and perhaps <b>10% of the use cases will be analyzed in detail</b> . → Use Case의 이름을 정도만 파악. |
| Supplementary Specification      | Describes other requirements, mostly <u>non-functional</u> . During inception, it is useful to have some idea of the key non-functional requirements that have will have a major impact on the architecture.          |
| Glossary 용어집.                    | Key domain terminology, and data dictionary.                                                                                                                                                                          |
| Risk List & Risk Management Plan | Describes the risks (business, technical, resource, schedule) and ideas for their mitigation or response.                                                                                                             |
| Prototypes and proof-of-concepts | To clarify the vision, and validate technical ideas.                                                                                                                                                                  |
| Iteration Plan                   | Describes what to do in the first elaboration iteration.                                                                                                                                                              |
| Phase Plan & SW Development Plan | Low-precision guess for elaboration phase duration and effort. Tools, people, education, and other resources.                                                                                                         |
| Development Case                 | A description of the customized UP steps and artifacts for this project. In the UP, one always customizes it for the project.                                                                                         |



# The UP Development Case (Reminder)

- 프로젝트에서 사용할 실천법( practices)과 산출물( Artifacts)의 선택을 보여주는 문서. → 프로젝트에 맞게 V를 커스터마이징.
- **Development Case:** ⇒ Shows the choice of practices and artifacts for a project
    - In fact, an artifact in the Environment discipline
    - (e.g.) the development case for the NextGen POS case study

| Discipline          | Practice                                                                           | Artifact<br>Iteration→      | Incep.<br>I1 | Elab.<br>E1..En | Const.<br>C1..Cn | Trans.<br>T1..T2 |
|---------------------|------------------------------------------------------------------------------------|-----------------------------|--------------|-----------------|------------------|------------------|
| Business Modeling   | agile modeling<br>req. workshop                                                    | Domain Model                |              | s               |                  |                  |
| Requirements        | req. workshop<br>vision box exercise<br>dot voting                                 | Use-Case Model              | s            | r               |                  |                  |
|                     |                                                                                    | Vision                      | s            | r               |                  |                  |
|                     |                                                                                    | Supplementary Specification | s            | r               |                  |                  |
|                     |                                                                                    | Glossary                    | s            | r               |                  |                  |
| Design              | agile modeling<br>test-driven dev.                                                 | Design Model                |              | s               | r                |                  |
|                     |                                                                                    | SW Architecture Document    |              | s               |                  |                  |
|                     |                                                                                    | Data Model                  |              | s               | r                |                  |
| Implementa-<br>tion | test-driven dev.<br>pair programming<br>continuous integration<br>coding standards | ...                         |              |                 |                  |                  |
| Project Management  | agile PM<br>daily Scrum meeting                                                    | ...                         |              |                 |                  |                  |
| ...                 |                                                                                    |                             |              |                 |                  |                  |



# How much UML during Inception?

- The purpose of inception is to collect just enough information to
  - establish a common **vision**
  - decide if moving forward is **feasible**
  - decide if the project is **worth serious investigation** in the elaboration phase.
- Much UML diagramming is not required.
  - Inception has more **focus on understanding the basic scope** and 10% of the requirements, expressed mostly in text forms.
  - In practice, most UML diagramming will occur in the next phase elaboration.

↳ inception scope와 feasibility 를 위주로 본다.  
UML 다이어그램은 Elaboration 단계에서 시작.



# Chapter 5

## Evolutionary Requirements

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# Requirements in UP

- Requirements
  - Capabilities and conditions to which the system must conform
- UP requirements are analyzed iteratively and skillfully.
- UP encourages skillful elicitation (finding) via techniques such as
  - writing use cases with customers,
  - requirements workshops that include both developers and customers,
  - a demo of the results of each iteration to the customers, to solicit feedback.



# Category of Requirements

- In the UP, requirements are categorized according to the **FURPS+** model
  - Functional : features, capabilities, security
  - Usability : human factors, help, documentation
  - Reliability : frequency of failure, recoverability, predictability
  - Performance : response times, throughput, accuracy, availability, resource usage
  - Supportability : adaptability, maintainability, internationalization, configurability
- **URPS** are called Quality Requirements F: functional requirement.

**Constraints** (the “+” in FURPS+) → *constraint*

- Implementation : resource limitations, languages and tools, hardware, ...
- Interface : constraints imposed by interfacing with external systems
- Operations : system management in its operational setting
- Packaging : for example a physical box
- Legal : Licensing and so forth



# Requirements Organization in UP

- Several requirements artifacts. (But, they are all optional)
  - **Use-Case Model** → functional requirement
    - A set of typical scenarios of using a system.
    - Primarily for functional (behavioral) requirements.
  - **Supplementary Specification** → quality requirement.
    - Basically, everything not in the use cases
    - Primarily for all non-functional requirements, such as performance or licensing.
    - Also place to record functional features not expressed (or expressible) as use cases
  - **Glossary** → 용어정의
    - Defines noteworthy terms.
    - Data dictionary recording requirements related to data, such as validation rules, acceptable values, and so forth.
  - **Vision**
    - Summarizes high-level requirements.
    - A short overview document for quickly learning the project's big ideas.
  - **Business Rules**
    - Requirements or policies.
    - required in the domain or business (e.g., government tax law)

규칙/정책



# UP Artifacts Relationships

